

UNITED STATES SODIUM COMPONENTS PROGRAMME

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1. Introduction

Development of sodium components in the United States has been conducted primarily by the laboratories and industrial organizations involved in the design, fabrication, construction, or operation of the major U.S. sodium cooled reactor plants, such as the Experimental Breeder Reactor-II (EBR-II), the Sodium Reactor Experiment (SRE), the Hallam Nuclear Power Facility (HNPF), and the Enrico Fermi Atomic Power Plant (EFAPP). As a result of design and operating experiences with these plants, sodium components are being both proved and developed, and requirements for additional component development efforts and test facilities are being identified.

The AEC initiated a general sodium components effort in 1958, known as the Sodium Components Development Programme. The primary objective of this effort was to develop heat transport system components for future sodium cooled nuclear plants that would be designed to operate at a maximum reactor sodium outlet temperature of 650°C. Reactor plant operating experience and results of the Sodium Components Development Programme are currently being reviewed with the objective of establishing a components programme capable of meeting the requirements of the Liquid Metal Fast Breeder Reactor (LMFBR) Programme. In establishing the LMFBR component programme, the AEC will continue to utilize the most qualified laboratories and industrial organizations.

This paper discusses the experience and current status of the Sodium Components Development Programme and some of the significant results of the evaluation of the operation of Hallam and SRE components since the ANS Topical Meeting on Fast Reactor Technology in April of 1965.⁽¹⁾

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Current operating experience at EBR-II and Fermi is included in other papers presented at this Conference.

2. Components Operation Evaluation—HNPF and SRE

Hallam Nuclear Power Facility

Prior to the graphite moderator can failure which shut down the Hallam 75 MW(e) reactor, the reactor had accumulated 16,000 operating hours. Initial difficulties and associated modifications to the pumps, intermediate heat exchangers, and steam generators have been previously reported.⁽¹⁾ Operation of these components after modification was essentially trouble free.

Post-operative examination indicated that a sodium leak occurred in one of the type 304 stainless steel secondary expansion tanks, apparently caused by corrosion from weld rod coating fluoride-bearing residue trapped in a void on the tank wall. Of equal interest as to the cause of the leak was the corrosion induced by the leaking sodium on the outside of the tank. Considerable wastage of stainless steel occurred under the thermal insulation. This corrosion was similar in appearance to that caused by Na leaks in mass transfer loops.

Post-operative evaluations indicate that hydrogen contamination may be the reason for the periodic increase in plugging temperature of the secondary loops; such contamination necessitated periodic cold trapping.⁽²⁾ This postulation is based on experiments conducted by the Detroit Edison Co. for the EFAPP steam generators. These experiments demonstrated that hydrogen, produced by corrosion of the $2\frac{1}{2}$ Cr-1Mo steel on the water side of the tube, diffused through the tube wall to the sodium side and resulted in increased hydrogen contamination levels. The qualitative nature of the results is associated with uncertainties in the corrosion rate and with conflicting data on hydrogen solubility in sodium. No deleterious effects have been detected on materials due to the apparent increase in contaminant levels; however, periodic cold trapping was employed to keep the plugging temperature down.

Sodium Reactor Experiment

The SRE has operated over 17,000 hours at temperatures at or above 454°C from 1957 to February 1964. The Empire States Atomic Development Associates (ESADA), a group of New York State investor-owned utilities, supported a detailed examination of system components removed during a modification programme instituted subsequent to February 1964. The most noteworthy conclusion was the lack of apparent effect of sodium operation.⁽³⁾ A thin (25 Micron) layer of hardened material at sodium-stainless steel interfaces was noted in many cases, which showed the same composition as the base material, but was work-hardened in structure. No cracking or

indication of incipient failure was evident. Of note, however, was the unchanged physical appearance of an eductor nozzle through which sodium was passed at velocities of 18 m/sec during the operating period (Fig. 1). Discoloration caused by a surface carbon deposit appears; there is no evidence of erosion or corrosion despite the high sodium velocity at a temperature of about 245°C.

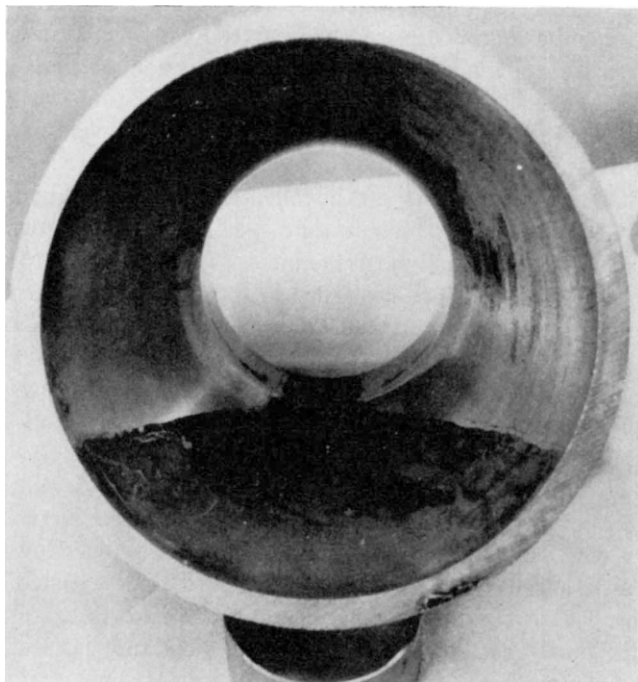


FIG. 1. Eductor nozzle after 18 m/sec sodium velocity test.

3. Large Sodium Component Design and Development

Steam Generator Development

The current steam generator design and development efforts by the USAEC include installation and test of the Alco Products-Baldwin-Lima-Hamilton steam generator in the Sodium Components Test Installation (SCTI), design and fabrication of the Atomics international modular steam generator now awaiting test and the design of the Babcock & Wilcox unit presently under fabrication.

Sodium Components Test Installation

The steam generator testing is conducted in the Sodium Components Test Installation (SCTI) which has been described in previous reports.⁽⁴⁾ The SCTI

is a 35 MW(th) test facility utilizing a natural-gas fired heater to heat a primary sodium circuit which transports the heat through an intermediate heat exchanger to a secondary sodium loop to the test steam generator. Start-up of this facility has uncovered several difficulties. The bellows of three small (50 mm) bellows sealed, globe-type throttling valves have failed, allowing small amounts of sodium to leak out of the body vent. These stainless steel valves were a commercial item, purchased in accordance with rigid specifications. Failure in the bellows apparently occurred due to flexing of the bellows plies. The cracks in the bellows were parallel to the convolutions, and near a point of attachment. Rather than embark on a bellows proof test programme these valves, as well as all other bellows sealed valves, were replaced with freeze stem sealed valves.⁽⁵⁾

Twelve large piping bellows are installed in the primary and secondary systems. These bellows were built to rigid specifications, were designed, and were analysed utilizing analytical methods developed by Atomics International.⁽⁶⁾ Three of these bellows failed in service. The leak stemmed from intergranular corrosion of the stainless steel, initiated by sensitization from a degassing heat treatment in manufacture. The heating was specified to eliminate gas between the two plies of the bellows. These bellows have been replaced, temporarily. A permanent modification to correct this difficulty is currently under study.

A surplus hermetically sealed, electric motor-driven pump is used to circulate secondary sodium. This pump had operated satisfactorily for over 1000 hours with either nitrogen or helium in the motor. However, when operated with argon the 4160 V motor experienced repeated insulation breakdowns. To correct this difficulty, caused by the low breakdown voltage of argon, the pump was modified by installing a rotating gas seal, similar to those of the Hallam Nuclear Power Facility pumps.⁽⁷⁾ This permitted motor operation in air which has resulted in satisfactory operation.

Test Programme

The test programme for each developmental steam generator is specified by the manufacturer, concurred in for practicability and safety by the test operator, and approved by the USAEC. Steam generator test data are being evaluated in two independent operations. Raw data are reduced by the operators, and supplied to both the steam generator manufacturer and an independent evaluation organization. Each organization then analyses and reports performance of the equipment and independently makes recommendations to the USAEC. The independent evaluation organization also reviews the test programme. It is anticipated that complete and impartial performance evaluation will thus be accomplished throughout the testing programme.

Alco-BLH Steam Generator

The currently installed Alco Products-Baldwin-Lima-Hamilton steam generator has already been described.⁽⁸⁾ It is a once-through, shell-and-tube type 316 stainless steel unit with Inconel cladding on the water side of tubes and tube sheets. Tubing flexibility is provided by sine-wave bends in the upper quarter of the tube length. It is instrumented with interior thermocouples on both sides of the tube sheets and along selected tubes on the sodium side, to measure temperature distribution and to infer thermal stresses. Strain gauges were used to measure strains during cold hydrostatic testing; results conformed to prediction at these conditions. Since strain gauges operable under sodium were not available, they were mounted on the exterior shell in the vicinity of the tube sheets. Thermal performance testing is currently under way at the SCTI.

Atomics International Modular Steam Generator

The AI modular steam generator⁽⁹⁾ now fabricated is awaiting test in the SCTI. Two complete evaporator-superheater module units, of 20 MW combined thermal capacity, are available. Fabrication of these units, with their butt welded tube-to-tube-sheet attachments designed to eliminate the crevices, was performed satisfactorily. Ultrasonic, dye check, and gammagraph inspections were passed. During fabrication, a section of tubing was accidentally damaged near the tube sheet. The 150 mm. damaged section was cut out, and a replacement section was welded into the tube using an inert gas tungsten electrode technique from inside the tube. This welding method was developed and used for attaching the tube to the tube sheet; its application to replacing sections of a tube was successful.

Babcock & Wilcox (B&W) Steam Generator

Conceptual design of the B&W 168 kg/cm² 566°C steam generator has been completed, and welding development is proceeding concurrently with initial fabrication of the prototype unit. Corrosion of 24Cr-1Mo by sodium hydroxide is being investigated. Analytical studies of sodium-water reactions are in progress. The conceptual design consists of separate superheater and evaporator tube bundles in the same sodium containing shell, with external connexions between the tube bundles to permit removal and replacement of bundles individually.⁽¹⁰⁾

ESADA Sodium-Water Reaction Programme

Empire States Atomic Development Associates is also supporting work at Atomics International to determine large sodium-water reaction effects in

TABLE 1. REACTION TEST RESULTS FROM STEAM-SIDE LEAK TO SODIUM

Test	Na (°C)	H ₂ O pressure (kg/cm ²)	Steam temperature (°C)	Steam pressure (kg/cm ²)	H ₂ O flow (g/sec)	Pressure at rupture (kg/cm ²)	Pressure 1.5 m above (kg/cm ²)	Pressure 1.5 m below (kg/cm ²)	Time per 1.5 m above (msec)	Time per 1.5 m below (msec)
0 ^a	—	—	560	105	—	—	—	—	—	—
1-A ^b	407	—	521	145	—	—	—	—	—	—
2-A	492	—	558	153	—	(56) ^c	28	26	0.6	0.6
1-B	493	192	—	—	5500	(105) ^d	27	41	0.6	0.6
2-B	482	235	—	—	5000	(—) ^e	24	24	—	—
3-B	492	192	—	—	5000	(240) ^f	11	34	—	—
1-C	489	240	—	—	5500	(43) ^g	21	29	0.75	0.65
2-C	493	244	—	—	5500	(34) ^h	19	24	0.90	0.90
3-C	489	242	—	—	5250	64	34	(—) ⁱ	0.40	(—) ⁱ
4-C	493	243	—	—	—	52	33	32	0.75	0.95

^a Preliminary—steam only—no sodium pressure in test section reached.^b Steam valve malfunction—no data.^c Peak pressure exceeded range of instrument (extrapolated value).^d Pressure due to impact.^e Premature rupture of burst diaphragm (failed-impact).^f (Impact) irregular pressure sequence.^g (Same instrument used for 3-B).^h Piezoelectric.ⁱ Failed.

the modular steam generator. One tube of a partial length 37-tube bundle is artificially ruptured; fast response instrumentation indicates the propagation of pressure waves, distribution of temperatures during the reaction, effects of geometry in placement of rupture disks, and displacement effects on tubing and stays. After reaction completion, tube bundles are disassembled for measurement and observation. The test apparatus (Fig. 2) supplies steam at 172 kg/cm^2 and 566°C , and solid water at 260°C and 211 kg/cm^2 , mocking up conditions in any part of the steam generator.⁽¹¹⁾ Results (Table 1) from ten reaction tests indicate that the reaction effects are containable.

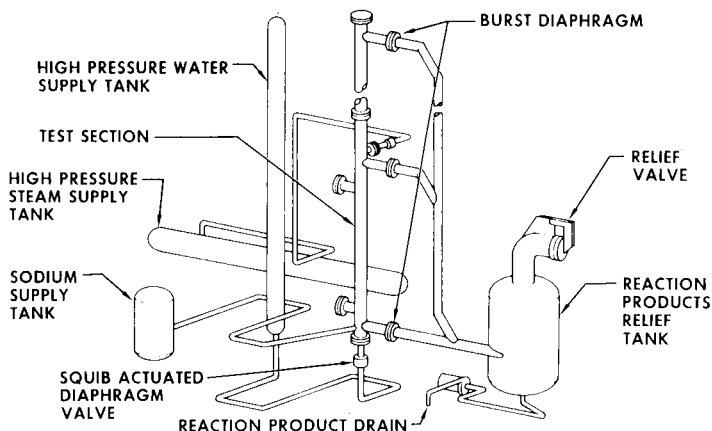


FIG. 2. Test apparatus for sodium-water reaction.

Sodium-Water Reaction Tests—Small Leaks

A small leak in one water tube of a sodium heated steam generator can cause the successive failure of adjacent tubes by the corrosion-erosion effect of the resulting sodium-water reaction. In order to obtain the information necessary to evaluate, and thus prevent or control this phenomenon, three experimental efforts have been conducted, and one effort is currently in progress.

The Detroit Edison Co. tests and the tests for the AEC-B&W steam generator were conducted in test devices similar to that shown schematically in Fig. 3. Reaction vessels ranged from 200 to 300 mm in diameter.

Detroit Edison Co. Tube Leakage Test for EFAPP—1962

The primary objective of this test was to determine if water leaks caused by stress-corrosion cracks would increase or cause damage to adjacent tubes. Two tubes containing stress-corrosion cracks were removed from a rejected

tube bundle of an EFAPP steam generator. The removed tubes were dead-ended and immersed in Na in the reaction vessel. Batches of 250 and 500 g of water were forced through the cracks at 67 kg/cm^2 over a period of 22 hours. Leakage through the tubes increased tenfold during the test; however, no damage to material surrounding the cracks was found nor was the microstructure of the $2\frac{1}{4}\text{Cr-1Mo}$ material altered. The maximum tube temperature attained was 445°C (a rise of about 165°C).

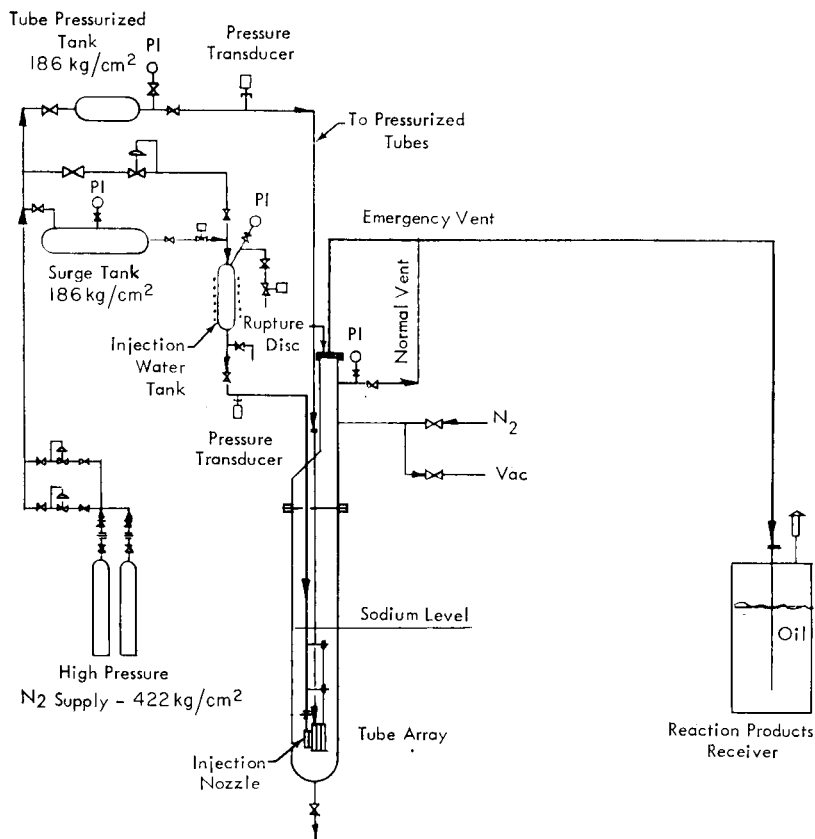


FIG. 3. Sodium-water tube thinning tests—schematic.

Detroit Edison Co. Tube Thinning Tests for EFAPP—1964⁽¹⁾

The purpose of these tests was to determine the possibility of a small water leak in the EFAPP steam generators causing thinning of a large number of tubes to the point of failure before the leak could be detected and appropriate action taken.

The reaction vessel and reaction products relief and collection system used in the test described above was also utilized for these tests. An injection nozzle

incorporating a metering orifice and rupture disk and connected to an argon pressurized water supply system was installed in place of the cracked tubes. The nozzle was directed at a group of nine dummy tubes submerged in sodium. The reaction was initiated by raising the water pressure to 910 kg/cm^2 , rupturing the rupture disk, and thus allowing water to flow through the orifice. Thermocouples were peened into the wall of several of the tubes and readings were recorded on a high response light beam oscillograph.

Eight tests were conducted in this series, with leak rates from 14 to 570 g/sec and time durations of 4 to 120 sec. Adjacent tube damage occurred for leak rates of 14 and 275 g/sec. The thinning was localized and wastage rates ranged from 5 to 50 microns/sec. No damage was evident at leak rates greater than 450 g/sec, this may have been due to the reaction products separating the reactants.

Tests of the leak detection system conducted in the EFAPP generators confirmed the fact that leaks could be detected before significant damage occurred to adjacent tubes.

Tube Thinning Test AEC-B&W Unit⁽¹²⁾

Tube thinning experiments were conducted to establish an upper limit for the number of tubes that could fail in the B&W unit before a leak was detected and corrective action initiated and accomplished before further serious damage could occur. These tests were conducted in a manner similar to that of the 1964 Detroit Edison Co. tests. The testing was accomplished in a unit similar to that mentioned above, with the exception that the reaction vessel had about six times the sodium inventory (34 kg) and water was injected at a pressure of 187 kg/cm^2 . Two tube arrangements were used; one arrangement simulated the boiler downcomer tubes and one simulated the boiler tube bundle. Leak rates ranged from 18 to 450 g/sec, and time durations ranged from 1.5 to 180 sec. The maximum wastage occurred at the 18 g/sec leak which persisted for 180 sec, and resulted in a loss of 0.6 mm. One test planned to be conducted at 450 g/sec also resulted in 0.7 mm loss of thickness. However, there is some evidence that a small leak may have occurred rather than the large leak planned due to incomplete fracture of the rupture disk. Two additional tests conducted at the 450 g/sec leak rate did not reproduce this result. Maximum reaction zone temperature of 1086°C occurred at a 450 g/sec leak rate.

Metallographic examination of attacked areas indicated uniform material removal rather than intergranular attack. Results of these tests indicated that an initial leak could cause an equivalent failure of about six tubes. Uncertainties associated with these tests and leak detection methods indicated that a significant safety factor should be applied to this number pending the results of the more general small leak programme discussed below.

General—Small Leak Sodium Reaction Test

The primary purpose of the small leak experimental programme is to determine the dominant parameters which effect tube thinning due to sodium–water reactions. Test conditions more nearly typify those which may occur in large LMFBR steam generators and, in particular, those of the B&W unit.

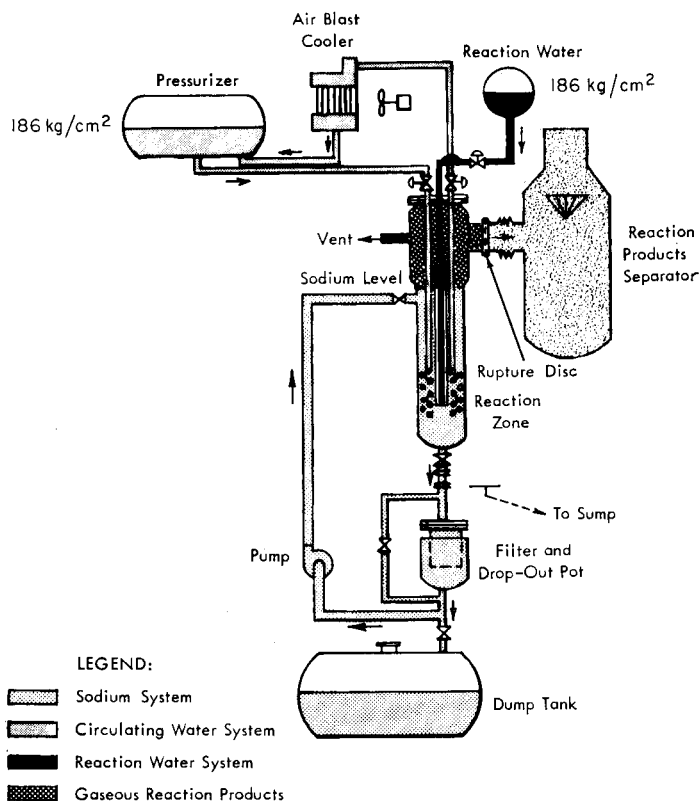


FIG. 4. Sodium–water reaction test—small leak.

Large quantities (3600 kg) of sodium are utilized in this facility to permit leaks to continue for long periods of time. Figure 4 is a schematic of the test apparatus. The variables that will be explored in this programme are leak rate, leak duration, tube spacing, sodium flow rate, and water flow rate. Leak rates ranging from 0.15 g/sec to 75 g/sec over time intervals of 5–10 min, will be investigated. The effects of these leaks will be established on dummy tubes of type 316 stainless steel and 2½Cr–1Mo steel, and a 2½Cr–1Mo steel tube containing circulating water.

Two tests have been conducted in this apparatus. The environmental

conditions for both of these tests were as follows: sodium flow rate, $109 \text{ m}^3/\text{h}$; bulk sodium temperature, 330°C ; sodium velocity, 0.4 m/sec ; water flow rate in target tube, 4.5 l/min ; injection water temperature, 330°C ; water injection rate, 1.3 g/sec ; injection duration, 300 sec . The first test utilized a flow baffle as the target. The injection point to target distance was 50 mm . This test resulted in a slight temperature increase and a loss of material in the target area of about 0.4 mm .

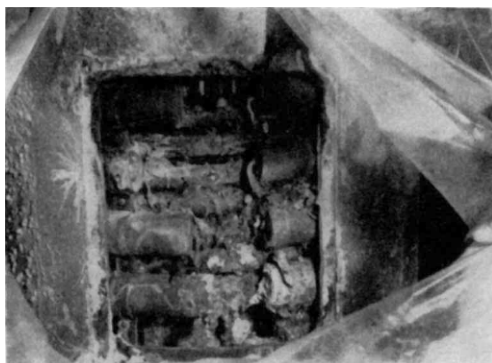


FIG. 5. General view of tube bundle assembly after test 2.



FIG. 6. Test 2 target tube (centre tube) after steam cleaning.

The second test injection was directed at a circulating water tube. The injection point to target distance was approximately 25 mm . During the 300 sec injection period, the sodium temperature in the target area increased approximately 110°C . At 320 sec after triggering, thermocouples indicated that additional reaction was occurring. Subsequent investigation indicated that the target tube had failed. Figures 5 and 6 show the target area before and after the cleaning. The failed tube was 25 mm in diameter, 3 mm wall thickness, and operated at 186 kg/cm^2 . Results of these tests are currently being analysed.

Sodium Pump Development

The sodium pump development efforts are based on the predicated long-term LMFBR programme requirements for 32,000 m³/h, 650°C pumps. Shaft sealed mechanical sump-type pumps are the prime candidate for fulfilling this requirement.

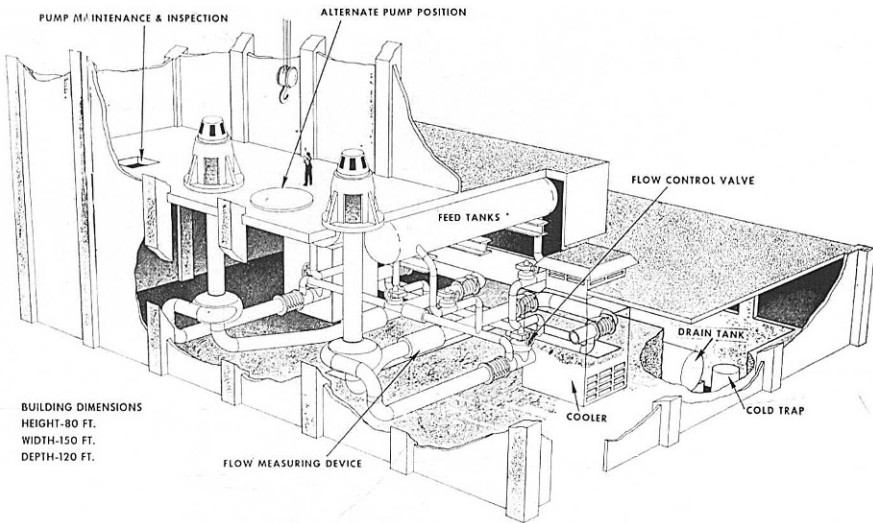


FIG. 7. Sodium pump test facility.

Sodium Pump Test Facility

A conceptual design study of a performance and proof-testing facility for sodium pumps was completed by Westinghouse in 1964. Planning for this facility is currently under way. The conceptual design is shown in Fig. 7. It is anticipated that this facility will accommodate performance, thermal shock, and endurance testing of pumps of up to 32,000 m³/h of sodium at 650°C. Flexibility for testing instrumentation in sodium (flow meters, pressure detectors, level probes), large sodium valves, and other components, is intended to be incorporated. The test facility will be able to proof test pumps prior to LMFBR plant installation.

Sodium Pump Design Study

Industrial organizations have been solicited by the AEC to produce designs of 16,000 m³/h shaft sealed, centrifugal, sump-type pumps that are five times greater in capacity than those now available. Some of the parameters which will be investigated in the designs are hot leg versus cold leg operation

and double versus single suction impellers. Pumps of this type will probably be installed in a large demonstration plant.

Sodium Pump Bearing and Seal Test

A programme is planned for testing full scale candidate bearings and seals for the 16,000 m³/h pump designs. The tests will determine performance characteristics, problem areas, and endurance of bearings and seals before construction of the 16,000 m³/h pumps. Test equipment will also accept bearings and seals for 32,000 m³/h pumps.

General Development Programmes

Precision mechanisms in sodium

The Precision Mechanism Programme is a basic programme which will provide engineering design data for fuel handling, control rod drives, and hold-down mechanisms. The initial effort is planned to determine compatible materials which can operate intermittently in sodium and in the cover gas over sodium. The tests will include elementary mechanical elements such as ball, roller and sleeve bearings, gears, slides, and grippers. Performance data and endurance of each of these elements will then be determined for reactor operating conditions. Results will be factored into improving the designs of the fuel handling and other components.

Large sodium valves

Initial investigation of the possible 1000 MW(e) LMFBR valve requirements, criteria, and concepts has been completed. Size requirements were found to be 700–750 mm check valves and 300–500 mm combination shut-off-flow control valves, and possibly 750 mm steam generator isolation valves. Conceptual designs for check, angle, and in-line applications were made.

Piping flexibility

A preliminary study of flexibility problems in large diameter, thin walled piping systems for reactor plants operating at 650°C has been completed. Both rigid pipe and expansion bellows design methods were examined and both were found inadequate for 650°C reactor plant applications. For rigid piping design, more rigorous design procedures are needed to augment the power piping code procedures which would (i) take full cognizance of fabrication and inspection procedures, (ii) assess the possibility of buckling caused by creep and/or loading, and (iii) recognize local overstrains and thermal ratcheting and their effects on elastic-plastic conditions.

4. Sodium Technology and Small Components

Mass Transfer

A study was initiated late in 1959 to evaluate candidate materials for sodium cooled reactors, and to develop means of predicting corrosion or mass-transfer rates in full-sized systems. Testing of type 316 stainless steel, $2\frac{1}{4}\text{Cr}-1\text{Mo}$ ($2\frac{1}{4}\text{Cr}$), and $5\text{Cr}-\frac{1}{2}\text{Mo}-\frac{1}{2}\text{Ti}$ (5Cr) alloy steels in six pumped test loops⁽¹⁴⁾ at various temperature conditions and oxide concentrations was completed in 1964. Loop material combinations are shown in Table 2.

TABLE 2. MATERIAL COMBINATIONS IN MASS TRANSFER TEST LOOPS

Loop no.	Hot-leg	Cold-leg	$T_{\max}$ (°C)
1	316 SS	316 SS	650
2	316 SS	$2\frac{1}{4}\text{Cr}$	650
3	316 SS	$2\frac{1}{4}\text{Cr}$	650
4	316 SS	5Cr	650
5	$2\frac{1}{4}\text{Cr}$	$2\frac{1}{4}\text{Cr}$	595
6	5Cr	5Cr	595

The statistical procedures and details of data analysis have been previously reported.^(15, 16)

Observations as to the effect of variables on weight change are as follows:

Oxide. The corrosion rates varied nearly linearly with sodium oxide impurity level.

Velocity. The corrosion rates varied nearly linearly with sodium velocity.

Time. The rates were independent of time after an initial period of several hundred hours. This period appears to correspond with establishment of a modified Fe-Ni-Cr composition surface layer.

Sample temperature. For each 56°C increase in temperature the corrosion rate was approximately tripled over a range of temperatures of 540–700°C.

Loop maximum temperature. Did not affect the local corrosion rate. This rate was controlled by the local temperature loop ΔT . No significant change in corrosion rate when varying ΔT between 107 and 246°C.

Specific transport effects reported⁽¹⁷⁾ involve the selective but not spatially localized transport of chromium and nickel in low oxide impurity runs (~ 10 ppm), selective transport of iron in high oxide impurity runs (~ 50 ppm), and transfer of carbon from $2\frac{1}{4}\text{Cr}-1\text{Mo}$ to type 316 stainless steel. Within the factor of 5 variation existing between experimental results, no statistically

significant difference in rates between type 316 stainless steel, $2\frac{1}{4}$ Cr, or 5Cr alloy steels was observed.

Since the experimental data are all based on a constant geometry or scale, this geometry effect has not been separated in this programme. To verify the postulated corrosion model and statistical equation, two corrosion mockup test loops simulating selected upstream regions in thermal, geometrical, and flow characteristics of a representative 1000 MW(e) design study plant have been designed. Loop scaledown factor is 1/30,000 (100 kW-6 g/m), chosen as a compromise between cost and corrosion sample sensitivity.

Mechanical Properties of Materials

Mechanical properties of $2\frac{1}{4}$ Cr-1Mo steel and type 316 stainless steel at 600–650°C in air, in helium, and in dynamic sodium having low or high oxygen or carbon content are being investigated⁽¹⁸⁾ and reported.^(19–21)

TABLE 3. TOTAL CARBON CHANGE OF 65-MIL SHEET SPECIMENS IN 4000 HOURS

Material	Carbon (ppm)		Carbon case thickness (mils)
	Original	Final	
Type 304 SS	667	9555	6.8
Type 316 SS	480	6115	5.6
$2\frac{1}{4}$ Cr-1Mo steel	981	3334	None

At 600°C, $2\frac{1}{4}$ Cr-1Mo steel showed no strong effects of sodium environment. Dynamic sodium caused decarburization and about 10% reductions in rupture strength, in life to 0.1% creep, and in tensile and yield strengths. Ductility was increased. In high oxygen sodium, the decarburization rate increased and surface cracking occurred, suggesting stress concentration sensitivity to sodium oxide content.

At 650°C, type 316 stainless steel shows subtle effects of environment. Rupture life is unaffected, fatigue life increases, and creep strength decreases about 15% in dynamic sodium. Tests in high oxygen or carbon content sodium are incomplete.

Metallographic, microhardness, chemical, and X-ray measurements of type 316 and type 304 stainless steels, Cr-Mo, and Armco Iron specimens exposed for a total of 4074 hours in carbon-contaminated dynamic sodium, reveal extensive surface carburization for both stainless steels, medium uniform carbon increase throughout $2\frac{1}{4}$ Cr-1Mo steel, and little change in Armco Iron. Carbon pickup is greater and penetrates more deeply and rapidly in type 304 than in type 316 stainless steel. Specimen surfaces include a very hard, high carbon case (see Table 3), and a subsurface transition zone. The

faster carburization of type 304 stainless steel is accompanied by more rapid transformation from austenite to ferrite and increased ferromagnetic behaviour over that observed in type 316 stainless steel. Carburization increased creep-rupture strengths but decreased ductility (see Table 4).

Sodium oxide effects on microstructure and composition of austenitic steels appear more complex than effects of carbon. Stainless steels types 304 and 316 oxidized differently in 300 ppm oxygen, 650°C sodium, than in 650°C air; X-ray examinations showed a discontinuous oxide layer of $\text{Na}_x\text{M}_y\text{O}_z$; microprobe results indicated the layer to be sodium chromite.

TABLE 4. COMPARISON OF DUCTILITY OF TYPE 316 STAINLESS STEEL DURING CREEP-RUPTURE TESTS IN LOW OXYGEN AND HIGH CARBON SODIUM

Stress lb/in ²	Elongation (%)		Reduction in area (%)	
	Low oxygen Na	High carbon Na	Low oxygen Na	High carbon Na
27,500	74	37	50	20
24,000	62	24	49	17
21,500	70	20	46	17
20,000	53	8	36	0

Extensive surface porosity was seen, but intergranular penetration of oxygen was slower than with carbon.

Sodium Technology Loop

A sodium loop is in operation for evaluating (i) in-line instruments to detect and determine impurities, (ii) purification techniques and devices, (iii) chemical and physical behaviour of impurities, and (iv) laboratory analytical techniques for contaminants. This loop⁽²²⁾ includes a 80 m³/h, 100 mm diameter main loop and four 2 m³/h, 25 mm by-pass circuits containing in-line instruments and purification devices.

Two rhometers provide inlet and outlet stream monitoring from any purification device, and a third rhometer can be placed parallel with either of these. Four United Nuclear Corporation oxygen meters and a plugging meter are installed to provide direct data comparisons. Chemistry samples are obtainable.

Purification devices planned for test include a cascade of three progressively finer micrometallic filters, two parallel hydroclones for 95% separation of 2 μ particles, a centrifuge, cold-traps and hot-traps.

Instrument responses to sodium carbonate, water, and sodium monoxide additions to loop sodium have been studied. Introductions of sodium carbonate into sodium⁽²³⁾ were unsuccessful; the solubility of sodium carbonate in sodium is being investigated separately. Eleven 5 ml water injections into

320,000 g of 230–355°C sodium were made over a 2 week period,⁽²⁴⁾ adding about 154 ppm oxygen to the initial, analytically determined oxygen concentration of about 70 ppm. Chemical analysis showed a decrease to 50 ppm

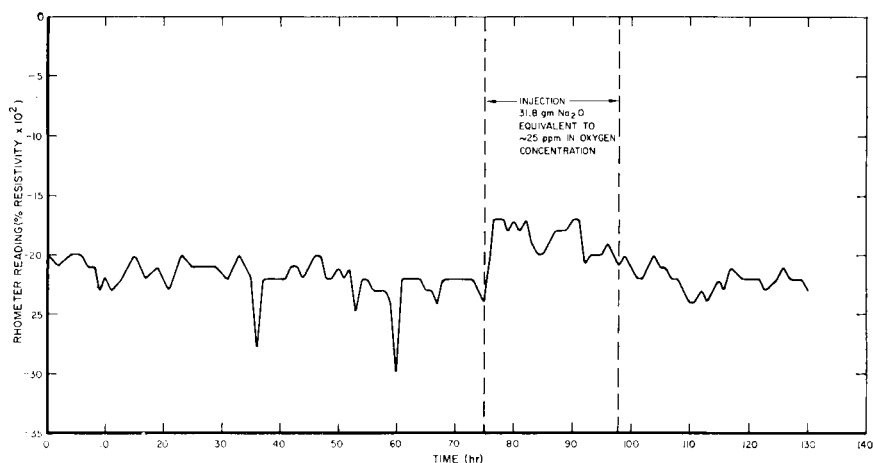


FIG. 8. Impurity concentration from sodium monoxide injection (oxygen rhometer).

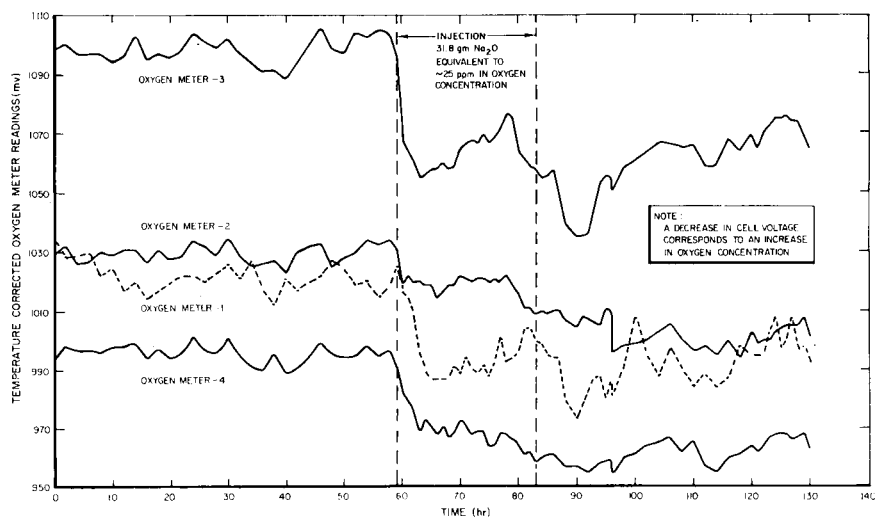


FIG. 9. Impurity concentration from sodium monoxide injection (oxygen meter).

instead of the expected increase to 224 ppm; also, the oxygen meters did not respond. The rhometers showed increases in sodium resistivity immediately after each injection. It is postulated that sodium hydroxide may have been formed. Two 31.8 g sodium monoxide injections were made into 300°C sodium, increasing oxygen content by about 50 ppm, and were registered on

rhometers and oxygen meters as shown in Figs. 8 and 9. Completion of sodium monoxide solution was reached in approximately 4 hours. Causes and corrections for fluctuations in meter responses are being sought. Temperature compensation difficulties⁽²⁵⁾ encountered in the Mark 1 rhometers are being corrected.

A hydrogen meter sensitive to 0.1 ppm, using bimetallic nickel-palladium tube exposed to flowing oxygen is being developed. This meter is based on the principle that hydrogen in the sodium will diffuse into an oxygen stream, and be converted to water for direct measurement by moisture meters. Experiments substituting argon for sodium indicated positive responses at 400°C for 500 ppm hydrogen.

Chemical Analysis of Sodium

AEC-sponsored round-robin analyses⁽²⁶⁾ for carbon in sodium and NASA-sponsored round-robins for oxygen in sodium and potassium all showed variations between specimens, and within and between participating laboratories. Research for improved analytic techniques is extensive, but beyond the scope of this paper.

Favourable results include development of amalgamation analysis for 1–5 ppm oxygen in sodium,⁽²⁷⁾ acceptable correlation of plugging and amalgamation data, and excellent correlations of oxide meter and distillation data. Investigations continue on gettering-vacuum fusion, 14 MeV neutron activation, melting point depression, coulometry and electrical resistivity for oxygen analysis, and wet and dry oxidation techniques for carbon analysis.

Liquid Metal Oxygen Meter

An in-line liquid metal oxygen meter⁽²⁵⁾ has been developed, providing continuous measurement of oxygen and rapid response to small changes in concentration, from 1 to 100 ppm. Response is insensitive to liquid metal velocity and temperature changes. Preferred electrode operating temperature is $315 \pm 30^\circ\text{C}$.

A sealed helium-filled, thoria-yttria electrode tube electrode assembly contains a copper-copper oxide reference electrode at the bottom. The bottom of this tube is exposed to flowing liquid metal, forming a galvanic cell across the thoria-yttria wall between the copper-copper oxide and sodium-sodium oxide electrodes. Since thoria-yttria electrolyte will pass only oxygen ions, the voltage developed by the cell is proportional to the difference in oxygen activities of the two electrodes and consequently is a direct measure of the effective oxygen concentration in the liquid metal.

A recording indicator contains a high input impedance differential voltmeter which measures the cell voltage; electrical features are incorporated which prevent transfer of oxygen ions through the electrolyte tube.

Twenty-one meters have been operated in forced circulation sodium loops. The instrument has also been tested satisfactorily in potassium.

The meters respond reproducibly to change in sodium-oxygen content with a linear relationship between meter voltage and the logarithm of oxygen concentration (Fig. 10) as determined by an integral sampler and vacuum distillation apparatus.

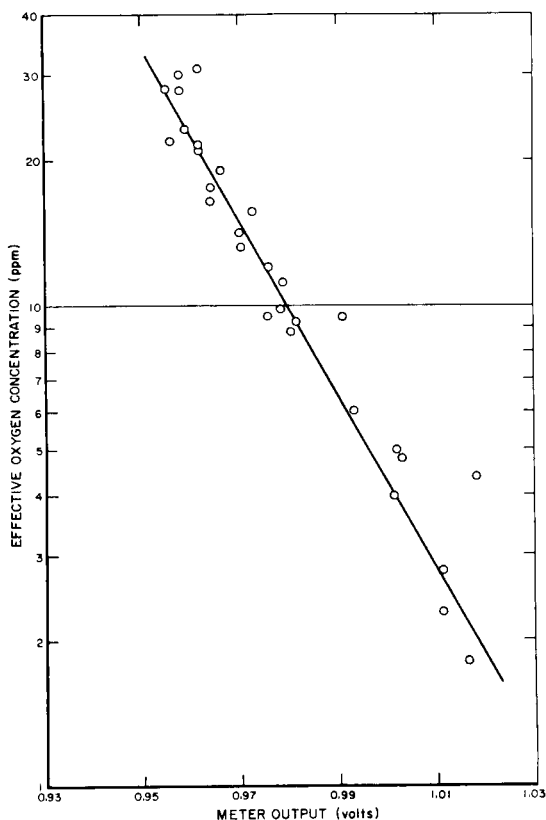


FIG. 10. Oxygen concentration versus meter voltage.

Experience with Plugging Indicators

Plugging meter designs vary widely; however, the American Society for Testing Materials is developing a standard plugging meter specification for corrosion tests. Plugging indications are usually consistent with cold-trap outlet temperatures, sometimes agreeing with chemical analyses, and are frequently confounded by multiple plugging indications. Multiple plugging indications have been observed by MSAR⁽²⁸⁾ and GE⁽²⁹⁾ with sodium, and by ORNL⁽³⁰⁾ with NaK. Plugging indicator responses to carbon content

have been observed at GE and MSAR. At ORNL, the NaK plugging indicator was sufficiently sensitive to detect the differential temperature solubility of argon and to reveal an unknown contaminant with plugging tendencies at about 275°C while cold-trapping below 150°C. Only mixed oxides of potassium and sodium were identified chemically. Similar observations at MSAR and GE also remain unidentified.

In-line Carbon Meter

A programme has been initiated to develop a continuous monitoring device for carbon activity in sodium. This programme is based on the concept of a diffusion type meter wherein carbon in sodium will be allowed to diffuse through a body-centred cubic iron membrane and be picked up by and reacted with a low pressure hydrogen gas stream. The resulting methane will be measured by either a flame ionization-hydrocarbon detector, or another suitable device.

Under-sodium Scanner

The feasibility of using directional ultrasonic beams for viewing and locating objects under sodium is being investigated⁽³¹⁾ using pulsed or frequency modulated ultrasonics.

Simple laws of reflection and refraction apply when the object boundary is large compared with wavelength. If the object is large compared with the wavelength, a shadow zone is formed behind the object. Selection of proper wavelengths for different dimensions of the object depends on scattering effects.

For high resolution in a single transducer scanner the object should be illuminated with a very fine beam. Horn type transducers are being investigated experimentally. A 2 mc/sec transducer using a combination of conical and wedge-shaped horns was successful in water.

Throat and mouth apertures of the wedge-shaped horn, dimension of the conical horn, and separation between the conical horn and wedge-shaped horn are important factors for obtaining adequate patterns. Image-sharpening circuits may be required in the receiving system. Ultrasonic-pulse techniques are preferred for single, narrow beam scanning.

Target range will be obtained from the oscilloscope separation between the transmitted pulse and the echo. When the transducer is rotated rapidly, the horizontal and vertical axes on the oscilloscope will represent the distance and the direction, respectively. The line-by-line image will be printed on the paper of a recorder. Ultimately electronic techniques will be used to transform the acoustic images into visible ones.⁽³²⁾

5. Summary

The experience to date on the operation of sodium system components has indicated that their safety and reliability is dependent on the degree to which the designer, fabricator, installer, and operator has established criteria, standards, specifications, quality control, and tests. Lack of *a priori* testing in prototypical environments has in many cases required that such proof tests be done in the reactor plant. Ensuing problems have delayed successful plant operation. In order to meet the objectives of the Liquid Metal Fast Breeder Reactor Programme, future performance of components must be measured against comparable components in high reliability, economic large scale power plants (fossil and fissile fired). A careful evaluation of the experience accumulated to date should assist in the formulation of a strong Sodium Components Programme. The establishment of component test centres should provide for a strong base on which to adequately develop components.

The successful development of components, the participation of industry, and the build-up of industrial capability will be necessary to provide the degree of confidence which will be needed to install safe and reliable components in the demonstration plant being considered. The base R and D programme now being established will provide the structure which should assure the U.S.A. that the components in the demonstration plant will operate safely and reliably.

The experience accumulated to date in the area of sodium technology indicates a need for a strengthened and augmented programme to provide for the development of sodium purification equipment, analytical techniques, and means of on-line detection of sodium and cover gas impurities to a level comparable with water technology.

References

1. J. J. MORABITO and H. W. SAVAGE, *Major Components and Test Facilities for Sodium Systems*, ANS-100, ANS National Topical Meeting on Fast Reactor Technology, Detroit, Michigan (1965).
2. D. K. DARLEY *et al.*, *Hallam Nuclear Power Facility Reactor Operations Analysis Program*, Monthly Report No. 25 (Sept. 1964).
3. D. T. EGGEN, *ESADA SGR Development Program*, Quarterly Progress Report, Oct.-Dec. 1964, AI-65-5 (1964).
4. J. A. LEPPARD, *Sodium Components Test Installation System Description*, NAA-SR-11671 (1966).
5. J. S. McDONALD, *Valve Stem Freeze Seal for High Temperature Sodium*, NAA-SR-4869 (1960).
6. W. F. ANDERSON, *Simplified Method of Bellows Analysis*, NAA-SR-9848 (1964).
7. R. ATZ, *Operating Experience with Heat Transfer System Pumps at the HNPF*, NAA-SR-9717 (1964).
8. *Final Design of 30 Megawatt Heat Exchanger and Steam Generator for Sodium Cooled Reactor System*, Alco Products Report APAE-112, Vols. 1-4 (1962).

9. D. F. CASEY, *Final Design of Sodium Heated Modular Steam Generators for the SCTI*, NAA-SR-9826 (1965).
10. *Preliminary Design Report, Prototype Steam Generator*, Babcock & Wilcox Report, B&W 1280-28 (1965).
11. *ESADA SGR Development Program*, Quarterly Progress Report, June 1965, AI-65-133 (1965).
12. W. L. CHASE *et al.*, *Analysis of the Babcock and Wilcox Full-size Steam Generator for a Sodium-Water Reaction*, APDA-176.
13. *Quarterly Technical Progress Report on AEC-sponsored Activities*, AEC Contract No. AT911-10-865, APDA-174, Apr.-June 1965, and APDA-178, July-Sept. 1965.
14. R. W. LOCKHART, C. BILLURIS and M. R. LANE, *Sodium Mass Transfer: I. Test Loop Design*, GEAP-3725 (June 1962).
15. E. L. DUNN, *Sodium Mass Transfer: VI. Statistical Correlation of 1961-1962 Corrosion Data*, GEAP-4183 (Feb. 1964).
16. E. L. DUNN, K. B. STEWART and J. L. JAECH, *Sodium Mass Transfer: XIV. Statistical Analysis of 1961-1964 Sample Weight Change Data*, GEAP-4830 (Feb. 1965).
17. M. C. RAWLAND, D. E. PLUMLEE and R. S. YOUNG, *Sodium Mass Transfer: XV. Behavior of Selected Steels Exposed in Flowing Sodium Test Loops*, GEAP-4831 (Mar. 1965).
18. R. C. ANDREWS *et al.*, Topical Report No. 1, *Test Facility Design and Operational Procedures*, MSAR 63-161 (1963).
19. R. C. ANDREWS *et al.*, Topical Report No. 2, *Results of Physical Property Tests of 316 SS Specimens in 1200°F Sodium with Low Oxygen*, MSAR 64-18 (1964).
20. R. C. ANDREWS *et al.*, Quarterly Progress Report (Apr.-May-June 1965), *Effect of High Temperature Sodium on Austenitic and Ferritic Steels*, MSAR 65-96 (1965).
21. R. C. ANDREWS *et al.*, Quarterly Progress Report (July-Aug.-Sept. 1965), *Effect of High Temperature Sodium on Austenitic and Ferritic Steels*, MSAR 65-149 (1965).
22. *APDA 170 Quarterly Report*, p. 32 (Jan.-Mar. 1965).
23. *APDA 175 Quarterly Report*, p. 31 (Apr.-June 1965).
24. *Sodium Technology Loop, Evaluation of Inline Instruments and Purification Devices*, AEC SCPP Info mtg, June 15-17, 1965, Chicago.
25. B. MINUSHKIN, The liquid metal oxygen meter, *Conference on Applications of High Temperature Instrumentation to Liquid Metal Experiments*, ANL, UNC-SPLM-61 (Sept. 1965).
26. R. W. LOCKHART *et al.*, *Results of the USAEC Round-robin Analysis for Carbon Impurity in Sodium*, GEAP-4602 (Mar. 1964).
27. B. MINUSHKIN, *Experimental Determination of Impurities in Sodium*, NDA-2154-6 (1961).
28. USAEC Programme: *Mechanical Properties of Materials in High Temperature Sodium*, Mine Safety Appliance Research, Callery, Pennsylvania.
29. USAEC Programme: *Mass Transfer in High Temperature Sodium*, GE-APED, San Jose, California.
30. H. W. SAVAGE *et al.*, *SNAP 8 Corrosion Program Summary Report*, ORNL-3898 (Dec. 1965).
31. YING and SCOTT, *Concepts for General Purpose Under-sodium Scanning Systems*, APDA-179 (Nov. 8, 1965).
32. V. G. PROKHOROV, The problem of converting an ultrasonic image into a visible one, *Soviet-Physics-Acoustics*, **3**, 272 (1957).